

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
LIDAM Energy Group, CORE, UCLouvain

I am applying for this position because the question at its centre is the one I want to spend my doctorate on. An auction award price is not the support a project receives. Pre-qualification requirements, completion bonds, penalty levels and local content conditions are priced into a bid before any award price exists, and each of them is set by the design rather than by the market. Which support scheme delivers capacity at what cost is therefore a question about rules and the incentives they create, and it is open to both theory and measurement.

Energy is the field I chose to specialise in at Cologne, which teaches it through the Institute of Energy Economics. I took *Energy Markets and Regulation* with Jun.-Prof. Dr. Oliver Ruhnau, whose chair is in energy market design, and finished it with 1.3 on a scale where 1.0 is the maximum. My final seminar paper is a research design on carbon leakage from the EU Emissions Trading System, and its contribution is the treatment variable. I define the effective carbon cost as the allowance price attenuated by the sector-year free-allocation share, so that a sector holding free allocation against most of its emissions is recorded as facing almost none of the headline price. Activity thresholds, closure provisions and partial-cessation rules tie EU allocation to continued production, which is what makes that attenuation a statement about incentives.

What I bring on method is causal inference on staggered policy adoption. My master's thesis, graded 1.0, estimates the labour-market effects of a national scholarship programme in Brazil on a panel I built from administrative records, roughly five million linked employer-employee observations across 558 microregions from 2005 to 2019, in R through BigQuery. Estimated with Callaway and Sant'Anna (2021) and doubly-robust inference, the programme raises wages by 3.04 per cent in low-exposure regions, distinguishable from zero at the 10 per cent level, along a dose-response that is non-monotonic in exposure. European auction policy has that same structure. Member states introduced auctions on different dates and have revised penalties, bid bonds, realisation deadlines and technology lots on different national timetables since.

Of the four directions in the call I would take up pre-qualification with completion bonds, and local content requirements, because both govern what a bid means before it is awarded. Willems and Yu compare discriminatory and uniform-price auctions on short-run bidding and long-run investment, and a penalty regime that alters the map from bid to delivered capacity alters that comparison. Lété, Smeers and Papavasiliou (2026) work out what zonal pricing with market-based re-dispatch costs in efficiency. My theory training is in game theory and competition, advanced microeconomics with Prof. Dr. Axel Ockenfels at 1.3, alongside econometrics at 1.3. A group holding auction theory and empirical energy economics in one place is where I want to do this work, and I would be glad to discuss it with you.

Available from October 2026. French B1, German and Tunisian Arabic native, English IELTS Academic 8.0. EU citizen. Referees: Prof. Dr. Pia Pinger and Jun.-Prof. Dr. Oliver Ruhnau, contact details on the CV.

Baha Khammari · b.e.khammari@gmail.com · +49 1520 9016956